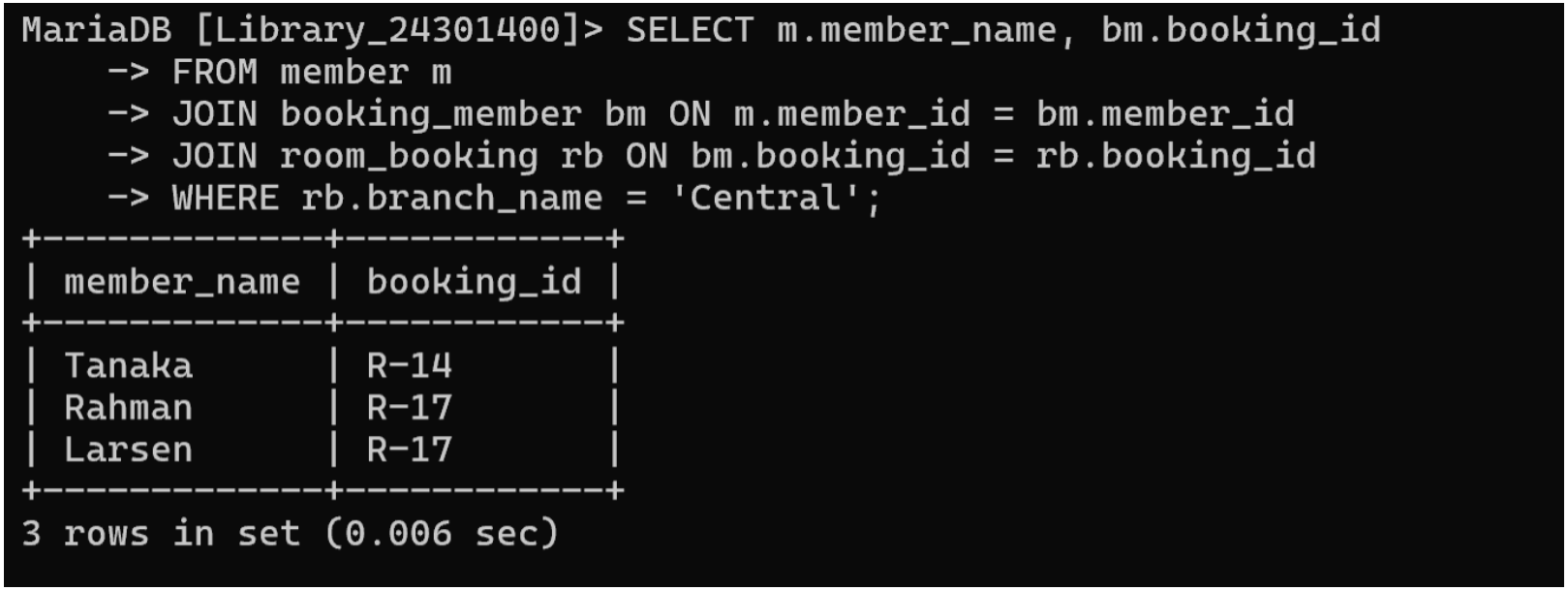
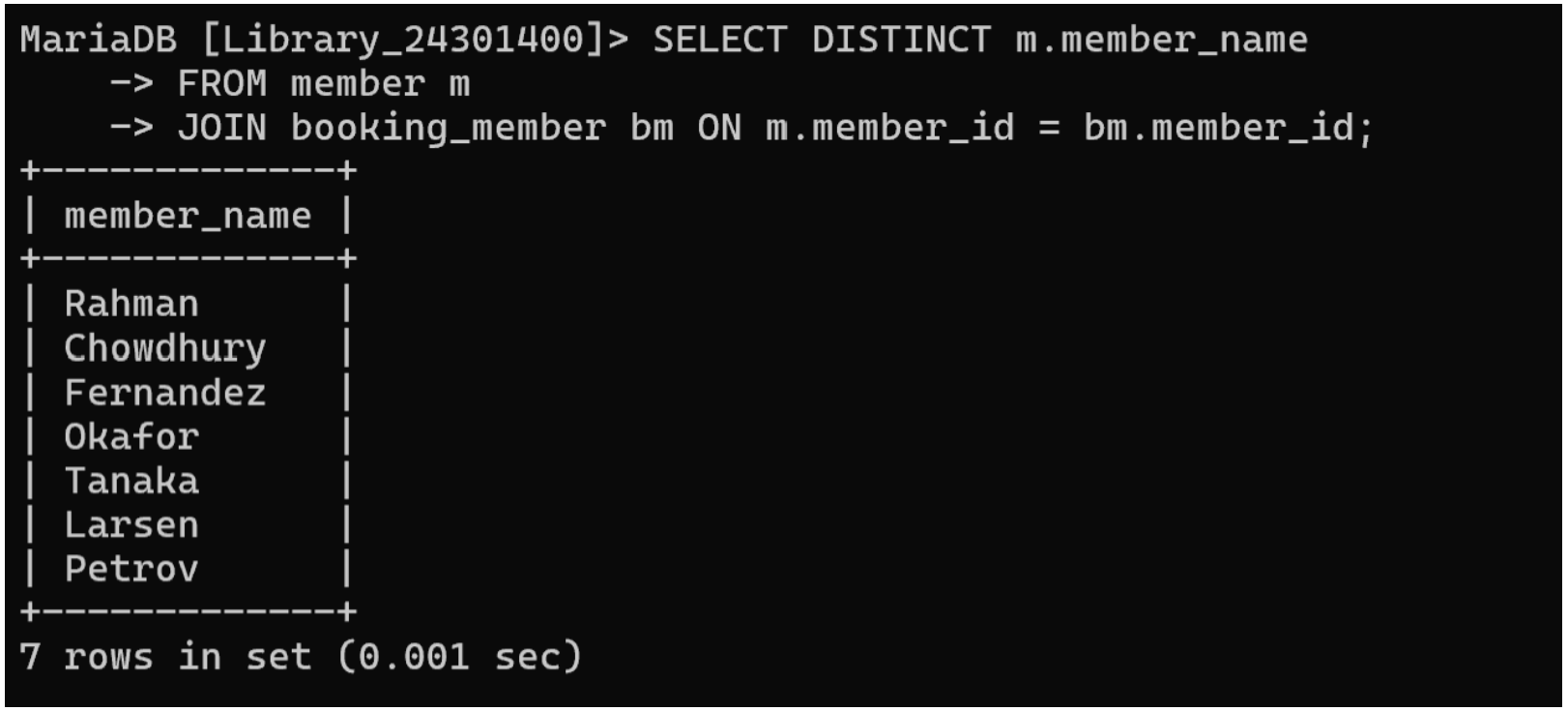
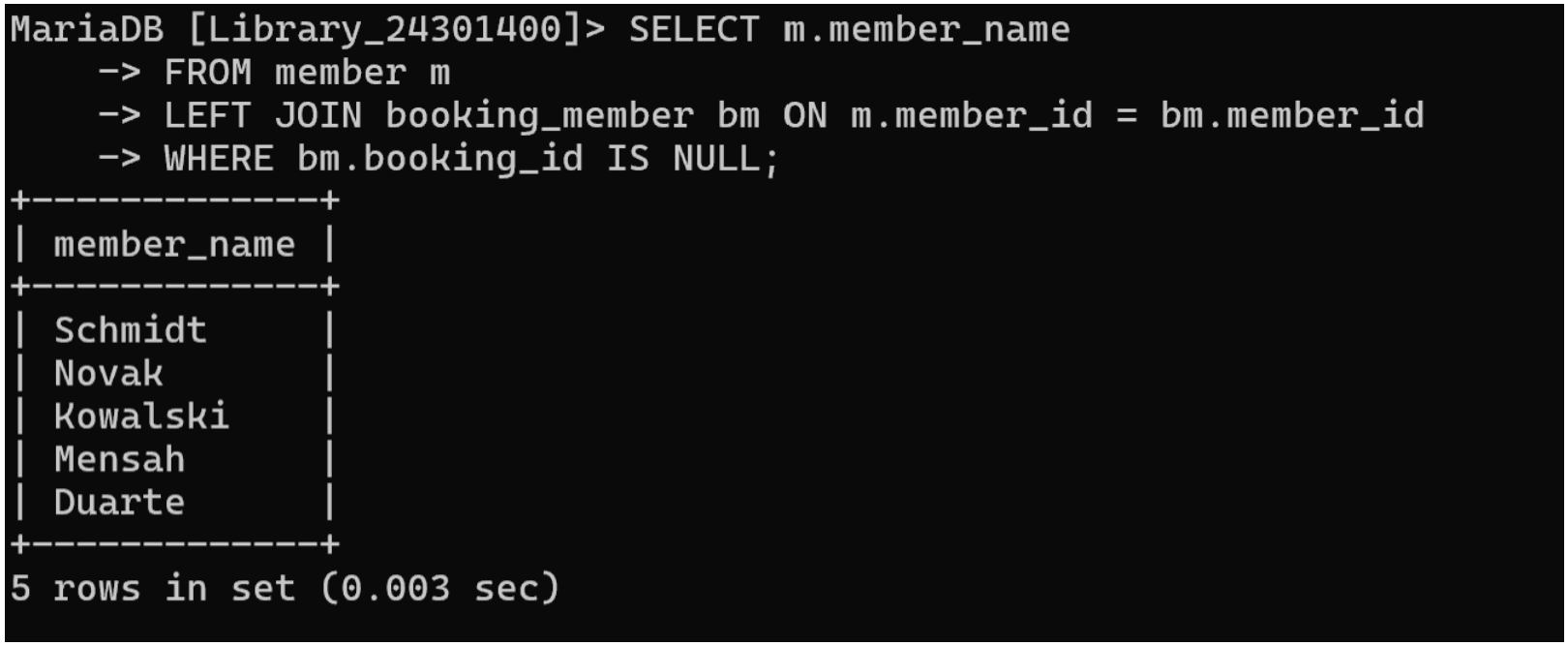
CSE370 : Database Systems

Assignment 01/02/03 | Fall 2026

ID : <24301400> | Name : Md. Mehtabul Islam

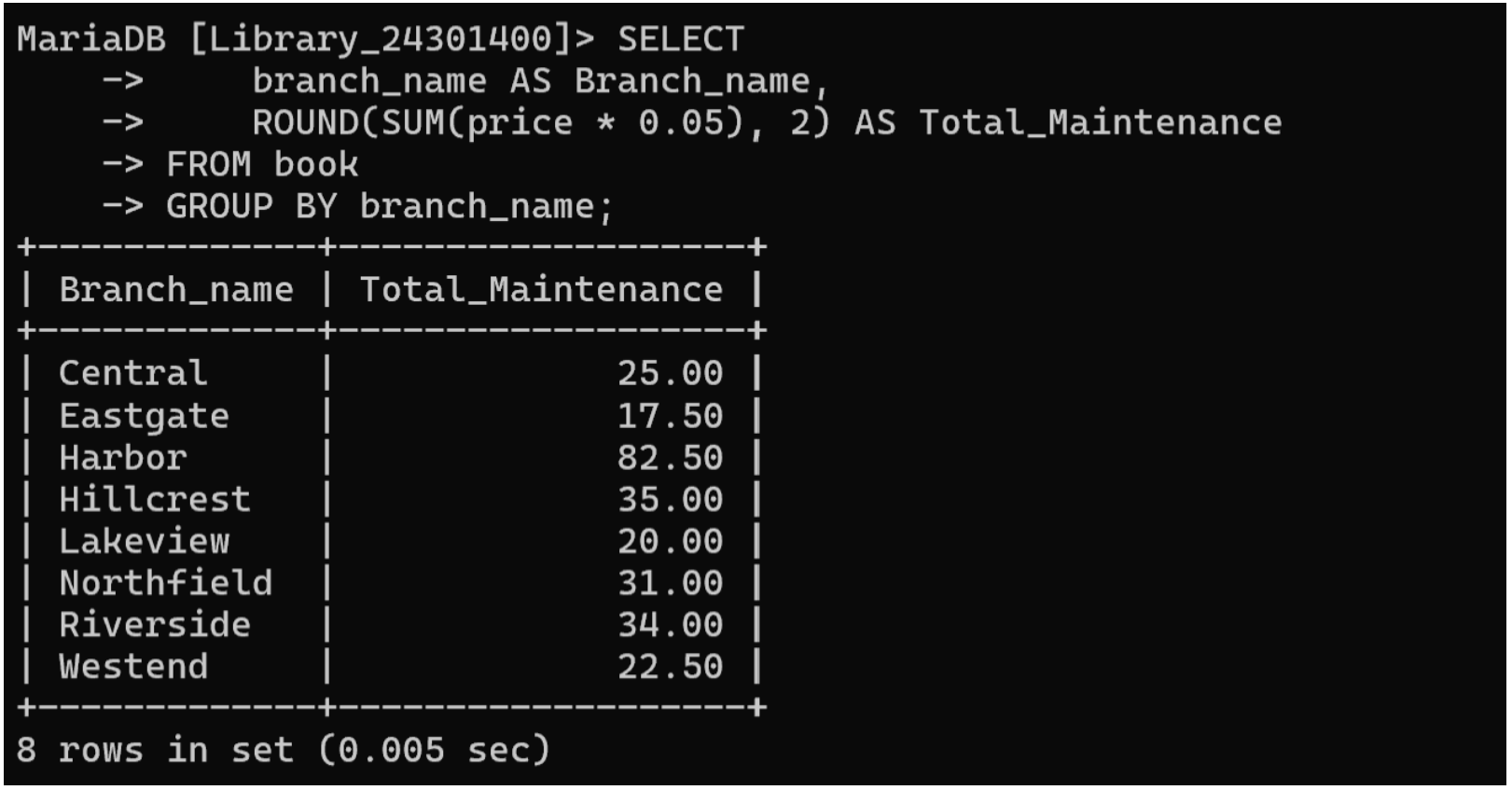
No 1 Query (as Plain Text)	SELECT m.member_name, bm.booking_id -> FROM member m -> JOIN booking_member bm ON m.member_id = bm.member_id -> JOIN room_booking rb ON bm.booking_id = rb.booking_id -> WHERE rb.branch_name = 'Central';
No 1 SS (of Query & Output in Shell)	
No 2 Query (as Plain Text)	SELECT DISTINCT m.member_name -> FROM member m -> JOIN booking_member bm ON m.member_id = bm.member_id;

No 2 SS **(of Query & Output in Shell)**	
No 3 Query **(as Plain Text)**	SELECT m.member_name -> FROM member m -> LEFT JOIN booking_member bm ON m.member_id = bm.member_id -> WHERE bm.booking_id IS NULL;

No 3 SS **(of Query & Output in Shell)**	
No 4 Query **(as Plain Text)**	SELECT -> m1.member_name AS Member1, -> m2.member_name AS Member2, -> m1.member_city AS City -> FROM member m1 -> JOIN member m2 -> ON m1.member_city = m2.member_city -> AND m1.member_id < m2.member_id;

No 4 SS
(of Query & Output in Shell)

```
MariaDB [Library_24301400]> SELECT
    ->     m1.member_name AS Member1,
    ->     m2.member_name AS Member2,
    ->     m1.member_city AS City
    -> FROM member m1
    -> JOIN member m2
    ->     ON m1.member_city = m2.member_city
    ->    AND m1.member_id < m2.member_id;
+-----------+-----------+-------------+
| Member1   | Member2   | City        |
+-----------+-----------+-------------+
| Rahman    | Fernandez | Springfield |
| Rahman    | Petrov    | Springfield |
| Chowdhury | Okafor    | Riverton    |
| Fernandez | Petrov    | Springfield |
| Tanaka    | Novak     | Oakdale     |
| Schmidt   | Kowalski  | Brookfield  |
| Larsen    | Duarte    | Maplewood   |
+-----------+-----------+-------------+
7 rows in set (0.002 sec)
```

No 5 Query (as Plain Text)	SELECT -> branch_name AS Branch_name, -> ROUND(SUM(price * 0.05), 2) AS Total_Maintenance -> FROM book -> GROUP BY branch_name;
No 5 SS (of Query & Output in Shell)	

No 6 Query (as Plain Text)	SELECT -> m.member_city, -> b.book_id, -> b.price -> FROM donor d -> JOIN member m ON d.member_id = m.member_id -> JOIN book b ON d.book_id = b.book_id -> WHERE (m.member_city, b.price) IN (-> SELECT -> m_inner.member_city, -> MAX(b_inner.price) -> FROM donor d_inner -> JOIN member m_inner ON d_inner.member_id = m_inner.member_id -> JOIN book b_inner ON d_inner.book_id = b_inner.book_id -> GROUP BY m_inner.member_city ->);

No 6 SS
(of Query & Output in Shell)

```
MariaDB [Library_24301400]> SELECT
    ->     m.member_city,
    ->     b.book_id,
    ->     b.price
    -> FROM donor d
    -> JOIN member m ON d.member_id = m.member_id
    -> JOIN book b ON d.book_id = b.book_id
    -> WHERE (m.member_city, b.price) IN (
    ->     SELECT
    ->         m_inner.member_city,
    ->         MAX(b_inner.price)
    ->     FROM donor d_inner
    ->     JOIN member m_inner ON d_inner.member_id = m_inner.member_id
    ->     JOIN book b_inner ON d_inner.book_id = b_inner.book_id
    ->     GROUP BY m_inner.member_city
    -> );
+-------------+---------+-------+
| member_city | book_id | price |
+-------------+---------+-------+
| Springfield | B-107   |   750 |
| Riverton    | B-102   |   700 |
| Oakdale     | B-105   |   900 |
| Brookfield  | B-109   |   450 |
+-------------+---------+-------+
4 rows in set (0.005 sec)
```

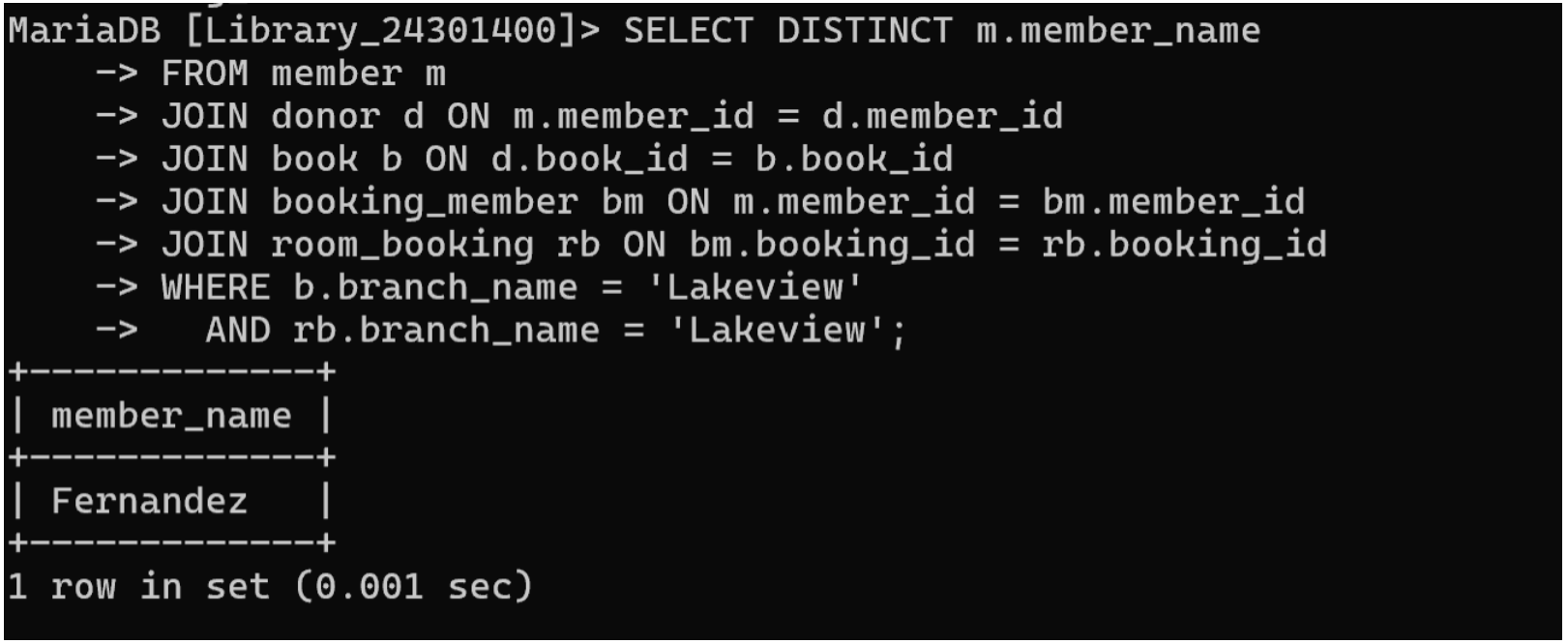
No 7 Query **(as Plain Text)**	SELECT RB.booking_id, RB.fee, M.member_name -> FROM Member M -> JOIN Booking_Member BM ON M.member_id=BM.member_id -> JOIN Room_Booking RB ON BM.booking_id=RB.booking_id -> JOIN (-> SELECT booking_id -> FROM Room_Booking -> ORDER BY fee DESC, booking_id DESC -> LIMIT 5 ->) AS TOP5 ON RB.booking_id=TOP5.booking_id -> ORDER BY RB.fee ASC, RB.booking_id DESC;

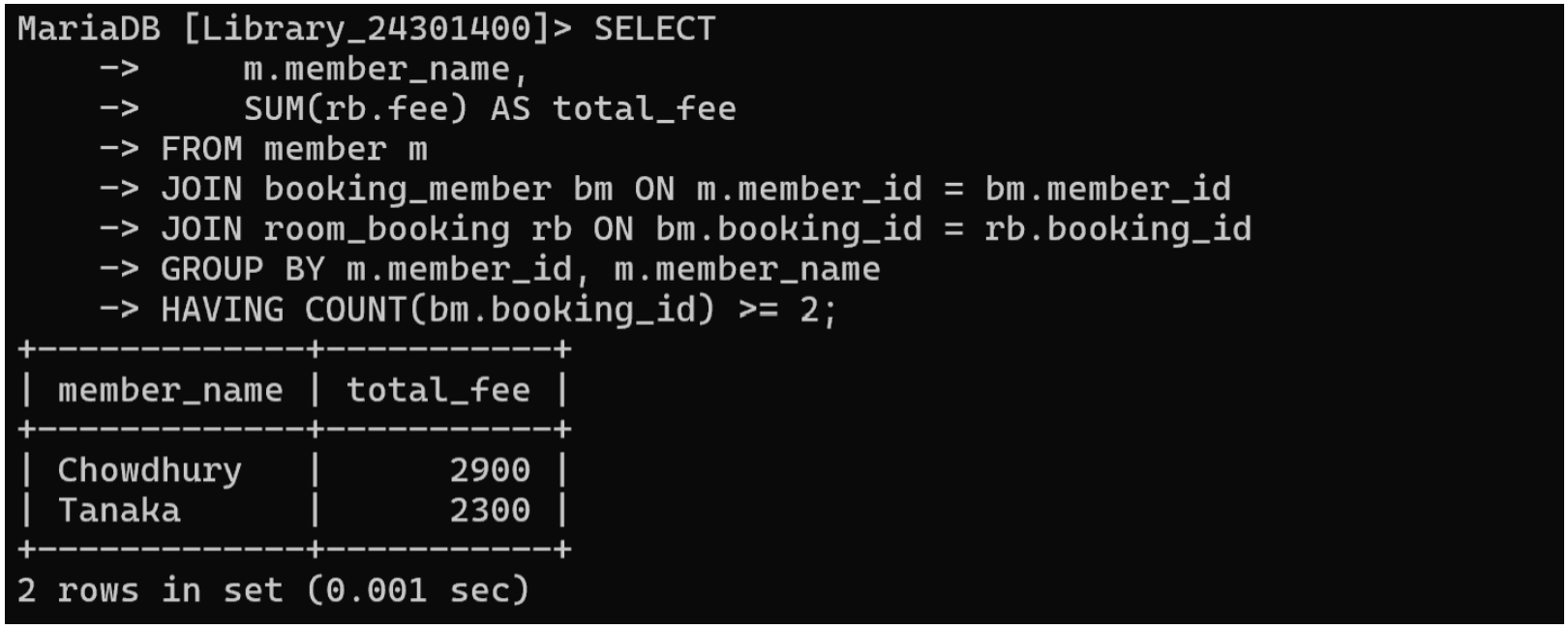
No 7 SS
(of Query & Output in Shell)

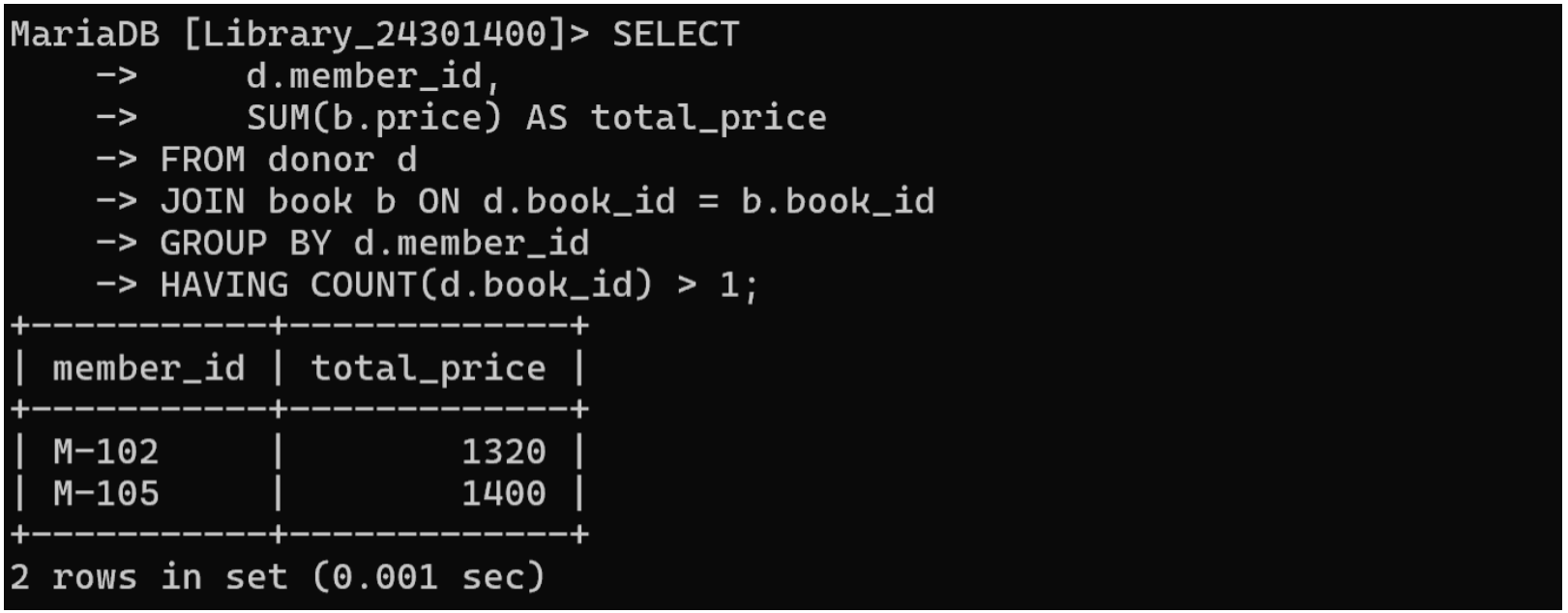
```
MariaDB [Library_24301400]> SELECT RB.booking_id, RB.fee, M.member_name
    -> FROM Member M
    -> JOIN Booking_Member BM ON M.member_id=BM.member_id
    -> JOIN Room_Booking RB ON BM.booking_id=RB.booking_id
    -> JOIN (
    ->     SELECT booking_id
    ->     FROM Room_Booking
    ->     ORDER BY fee DESC, booking_id DESC
    ->     LIMIT 5
    -> ) AS TOP5 ON RB.booking_id=TOP5.booking_id
    -> ORDER BY RB.fee ASC, RB.booking_id DESC;
+------------+------+-------------+
| booking_id | fee  | member_name |
+------------+------+-------------+
| R-11       |  900 | Chowdhury   |
| R-16       | 1300 | Petrov      |
| R-15       | 1500 | Fernandez   |
| R-14       | 1500 | Tanaka      |
| R-23       | 2000 | Chowdhury   |
+------------+------+-------------+
5 rows in set (0.005 sec)
```

No 8 Query
(as Plain Text)

```
SELECT DISTINCT m.member_name
  -> FROM member m
  -> JOIN donor d ON m.member_id = d.member_id
  -> JOIN book b ON d.book_id = b.book_id
  -> JOIN booking_member bm ON m.member_id = bm.member_id
  -> JOIN room_booking rb ON bm.booking_id = rb.booking_id
  -> WHERE b.branch_name = 'Lakeview'
```

	-> AND rb.branch_name = 'Lakeview';
No 8 SS (of Query & Output in Shell)	
No 9 Query (as Plain Text)	SELECT -> m.member_name, -> SUM(rb.fee) AS total_fee -> FROM member m -> JOIN booking_member bm ON m.member_id = bm.member_id -> JOIN room_booking rb ON bm.booking_id = rb.booking_id -> GROUP BY m.member_id, m.member_name -> HAVING COUNT(bm.booking_id) >= 2;

No 9 SS **(of Query & Output in Shell)**	
No 10 Query **(as Plain Text)**	SELECT -> d.member_id, -> SUM(b.price) AS total_price -> FROM donor d -> JOIN book b ON d.book_id = b.book_id -> GROUP BY d.member_id -> HAVING COUNT(d.book_id) > 1;

No 10 SS **(of Query & Output in Shell)**	

```
MariaDB [Library_24301400]> SELECT
    ->     d.member_id,
    ->     SUM(b.price) AS total_price
    -> FROM donor d
    -> JOIN book b ON d.book_id = b.book_id
    -> GROUP BY d.member_id
    -> HAVING COUNT(d.book_id) > 1;
+-----------+-------------+
| member_id | total_price |
+-----------+-------------+
| M-102     |        1320 |
| M-105     |        1400 |
+-----------+-------------+
2 rows in set (0.001 sec)
```